

TT Electronics

Product Lifecycle Process

Policy No: ENG 01

Document Owner:	Technology Director
Division / Department:	Engineering
Next Review Date:	Annually

Version History

Version	Date	Author	Reason
4.0	17th Nov 2025	Global Technology Director	Conversion of Gated Process Document (previously issued at V3) from guidance document format to TT standard process. Updated for changes to process flow agreed by Engineering Steering Group and reflecting process implemented in Cora.
4.1	4 th Mar 2026	Technology Director	Update to clarify and expand on NTI process (3.1.6) and align with new NTI gate checklist

Contents

1	INTRODUCTION	4
1.1	Policy	4
1.2	Scope	4
2	TT PRODUCT LIFECYCLE PROCESS	5
2.1	Phases and Gates	5
2.1.1	Project Phase	5
2.1.2	Gate Review	5
2.1.3	Phase 0 Project Initiation	5
2.1.4	Phase 1 Concept & Proposal	6
2.1.5	Phase 2 Requirements Development	6
2.1.6	Phase 3 Preliminary Design	6
2.1.7	Phase 4 Detail Design	6
2.1.8	Phase 5 Design Validation	7
2.1.9	Phase 6 Production Preparation & Qualification	7
2.1.10	Phase 7 Transfer & Monitor	7
2.1.11	Phase 8 Manufacture	8
2.1.12	Phase 9 End-of-Life	8
2.2	Alignment with Standards	9
3	PROJECT TYPES AND CATEGORIES	10
3.1	Project Types	10
3.1.1	NPI A (Large):	10
3.1.2	NPI B (Medium):	10
3.1.3	NPI C (Small):	10
3.1.4	NPI D Minor Customisations (Momentum)	10
3.1.5	Make-to-Print	10
3.1.6	NTI: New Technology Introduction	10
3.1.7	Continuous Improvement (CI) PDCA	12
3.1.8	Continuous Improvement (CI) DMAIC	12

3.1.9	Continuous Improvement (CI) TT BE Lean	12
3.2	Project Categorisation	12
3.2.1	Cat 1	12
3.2.2	Cat 2	13
3.2.3	Cat 3	13
4	GATE REVIEWS	14
4.1	Gate Content and Structure	14
5	PROJECT PORTFOLIO MANAGEMENT	15
5.1	Tools & Systems	15
5.2	Cora Functionality	16
5.2.1	Mandatory (Europe and North America Regions)	16
5.2.2	Recommended Functionality	17
6	GATE EXIT CRITERIA	19
6.1	Gate 0 Exit	19
6.2	Gate 1 Exit	20
6.3	Gate 2 Exit	20
6.4	Gate 3 Exit	20
6.5	Gate 4 Exit	20
6.6	Gate 5 Exit	20
6.7	Gate 6 Exit	20
6.8	Gate 7 Exit	21
6.9	Gate 8 Exit	21
6.10	Gate 9 Exit	21

1 Introduction

1.1 Policy

TT Electronics policy is to control and manage the development and introduction of all new products and technologies, whether TT Intellectual Property or not, within a common control framework, called the “TT Electronics Product Lifecycle Process”. This is a whole business process that encompasses all aspects of the project (i.e. meeting commercial, operational, technical, regulatory requirements and constraints) and all functions of the business shall be engaged.

This policy applies to all individuals working within the stages defined in scope for product and technology development, from scoping of the opportunity through to introduction into production, regardless of the type of the new product.

1.2 Scope

This document describes the TT Electronics Product Life Cycle Process with focus on the gated phases and reviews. It applies to;

- Products that are designed by TT for a specific customer (bespoke development),
- Products that are designed by TT for general sale,
- Products that are manufactured by TT to a customer design (make to print),
- Modifications to existing products that create new variants,
- Technologies that are developed by TT.

The project life cycle process applies to projects of all sizes and is defined in a way to scale from the smallest to the largest.

2 TT Product Lifecycle Process

2.1 Phases and Gates

The process for managing the lifecycle of a product is defined in terms of Phases, during which work is carried out, with a Gate Review between each to ensure the goals of the preceding Phase have been met before work begins in the next Phase. *(Note: at some locations within TT the term Stage may be used as a synonym for Phase, and the terms Stage Exit Review or Phase Exit Review may be used as synonyms for Gate Review).*

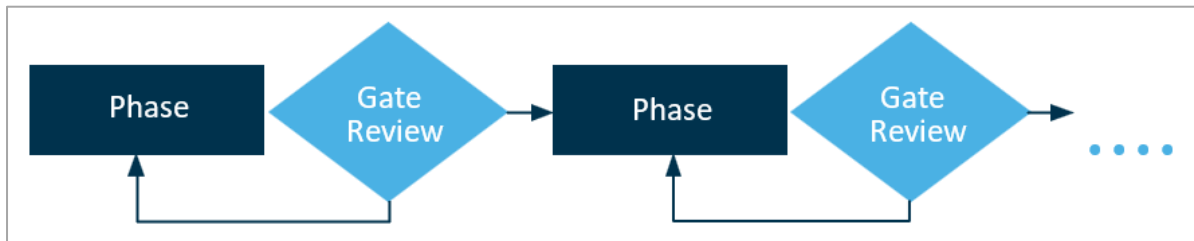


Figure 2.1.1 Phase and Gates Relation.

2.1.1 Project Phase

The project Phase is the part of the project where the tasks and deliverables which are required to meet the Gate criteria are completed.

2.1.2 Gate Review

A Gate Review is required at the end of each project Phase. If the gate review is passed, then the project moves to the next project phase.

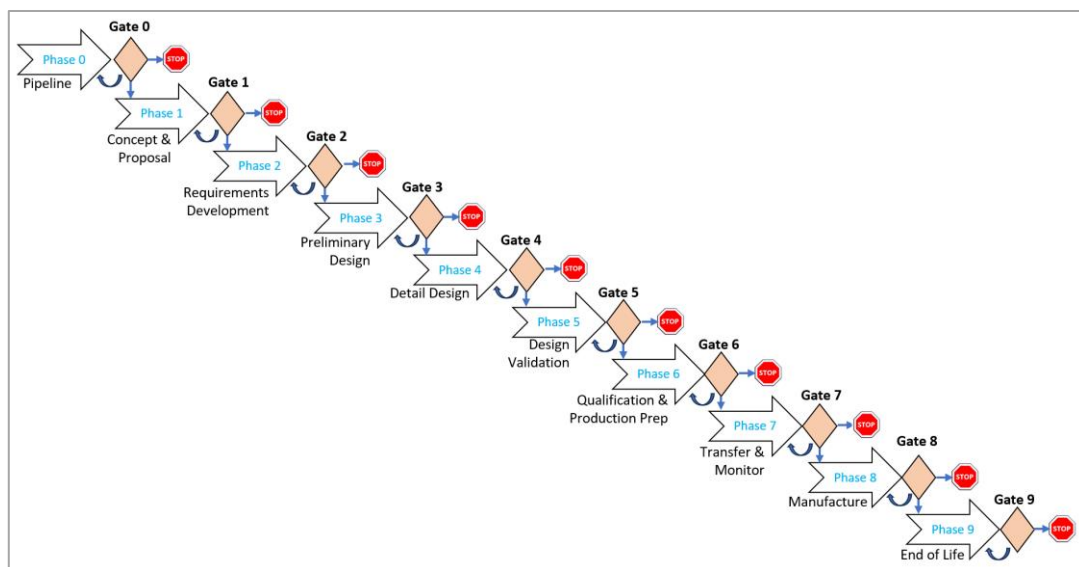


Figure 2.1.2 The Product Lifecycle Steps.

2.1.3 Phase 0 Project Initiation

New projects and opportunities are understood and assessed. The start is either triggered by a “Go” decision in a Go / No Go Review (within the Sales pipeline process) or from a

strategic product portfolio review that identifies an opportunity that requires internal investment.

2.1.3.1 Gate 0

Corresponds to the Sales Bid / No Bid review. The outcome of the Gate review is either a decision to do no further work or to continue to prepare a concept and costed proposal (proposal may be internal or external). Passing gate 0 is approval to start Phase 1.

2.1.4 Phase 1 Concept & Proposal

Analysis of requirements, develop conceptual design (unless project is make-to-print), confirm feasibility plus assess costs and times for development / introduction and recurring production. Output will include a costed proposal or costed business case.

2.1.4.1 Gate 1

Review outputs from concept and proposal phase, and ensure commercial, operational and technical aspects have been appropriately addressed. For externally funded projects, this is the approval of the proposal. For internally funded projects, approve the business case and funding. Exit from Phase 1. Passing gate 1 is approval to start Phase 2.

2.1.5 Phase 2 Requirements Development

Finalise project/product requirements. For projects including system development this will include both direct and derived requirements and may include both hardware and software where applicable.

2.1.5.1 Gate 2

Review to confirm that all necessary tasks and actions from the Requirements Development phase are completed and that the project is projected to meet critical requirements (schedule, cost, technical, quality, commercial, operational, etc). Passing gate 2 is approval to start Phase 3.

2.1.6 Phase 3 Preliminary Design

Develop the design including any necessary models, analysis and/or prototyping to validate the feasibility, performance, risks, costs and lead-times. May include functional prototyping to validate some aspects of the design. This phase will not normally apply to make-to-print projects.

2.1.6.1 Gate 3

Review to confirm that all necessary tasks from the Preliminary Design phase are satisfactorily complete, that actions from previous reviews are closed and that the project meets all critical requirements (schedule, cost, technical, quality, commercial, operational, etc). Passing gate 3 is approval to start Phase 4.

2.1.7 Phase 4 Detail Design

Mature the design detail sufficiently to allow validation parts to be built in the next phase, to complete any necessary models, analysis and/or prototyping to validate the feasibility,

performance, risks, costs and lead-times and to define required assembly, test and screening processes. Provisional production documentation (Bills of Material, routes, Standard Operating Procedures, Work Instructions, etc. as applicable) is generated. For make-to-print projects, tasks such as advising the customer on design for manufacture and designing test solutions, etc. may also be undertaken.

2.1.7.1 Gate 4

Review to confirm that actions from previous reviews have been addressed, all necessary tasks from the Detail Design phase are suitably completed and that the project continues to meet all critical requirements (schedule, cost, technical, quality, commercial, operational, etc). The design is considered “frozen” (version & configuration controlled) once this Gate is passed, although may be up-issued following feedback from later phases. Passing gate 4 is approval to start Phase 5.

2.1.8 Phase 5 Design Validation

During this phase the functional performance of the product is validated to ensure it meets design intent and requirements. Where applicable this may include formalised testing such as Design Assurance Test and pre-compliance environmental testing.

2.1.8.1 Gate 5

Review to confirm that actions from previous reviews have been addressed, all necessary tasks from the Design Validation phase are suitably completed and that the project continues to meet all critical requirements (schedule, cost, technical, quality, commercial, operational, etc). Passing gate 5 is approval to start Phase 6.

2.1.9 Phase 6 Production Preparation & Qualification

Finalise commissioning and validation of any new equipment in the production line if not already completed. Finalise all documentation needed to support production. A first production run may be completed to include applicable Qualification testing (note that for a “one-off” product, this may be the only production run).

2.1.9.1 Gate 6

Review to confirm that actions from previous reviews have been addressed, all necessary tasks from the Production Preparation and Qualification phase are suitably completed and that the project continues to meet all critical requirements (cost, technical, quality, commercial, operational, etc). Passing this gate approves the product to start full rate production. Outcome should include agreement on measure for completion of Phase 7. Passing gate 6 is approval to start Phase 7.

2.1.10 Phase 7 Transfer & Monitor

Production line run at rate, transfer to operations, start of production, lessons learned, continuous improvement, review of initial production to ensure technical and commercial requirements are being consistently maintained. For a repeating product produced in reasonable volume this phase will normally be a fixed amount of time, for a lower volume or product with irregular demand, this phase may cover the first 1-2 production runs. This needs to be agreed in Gate 6.

2.1.10.1 Gate 7

Review to confirm that actions from previous reviews have been addressed, and all necessary tasks from the Transfer & Monitor phase are suitably completed. Passing this gate marks the transfer of the product to operations (end of New Product Introduction). Passing gate is approval to start Phase 8.

2.1.11 Phase 8 Manufacture

Normal operations process, continuous improvement, financial monitoring, etc.

2.1.11.1 Gate 8

Kick-off of Phase 9. Identified need to obsolete a product, this Gate is held to agree the start of the end-of-life process (Phase 9) and may be part of the business Change Review Board or a similar change management process.

2.1.12 Phase 9 End-of-Life

Review options and business case to update or redevelop product. Iterate through this process from Gate 0 if a significant redevelopment is proposed. Plan for end-of-life, including last time buy policy, etc. Send notifications to customers and complete any final production. Update ERP for material status (obsolete, no longer available for purchase).

2.1.12.1 Gate 9

Review to confirm all end-of-life actions are complete. Finalises obsolescence of the product and marks end of production. Any product produced after this review must be subject to a separate documented business review, including consideration of all items normally addressed in Gate 6 before any Purchase Order can be accepted.

2.2 Alignment with Standards

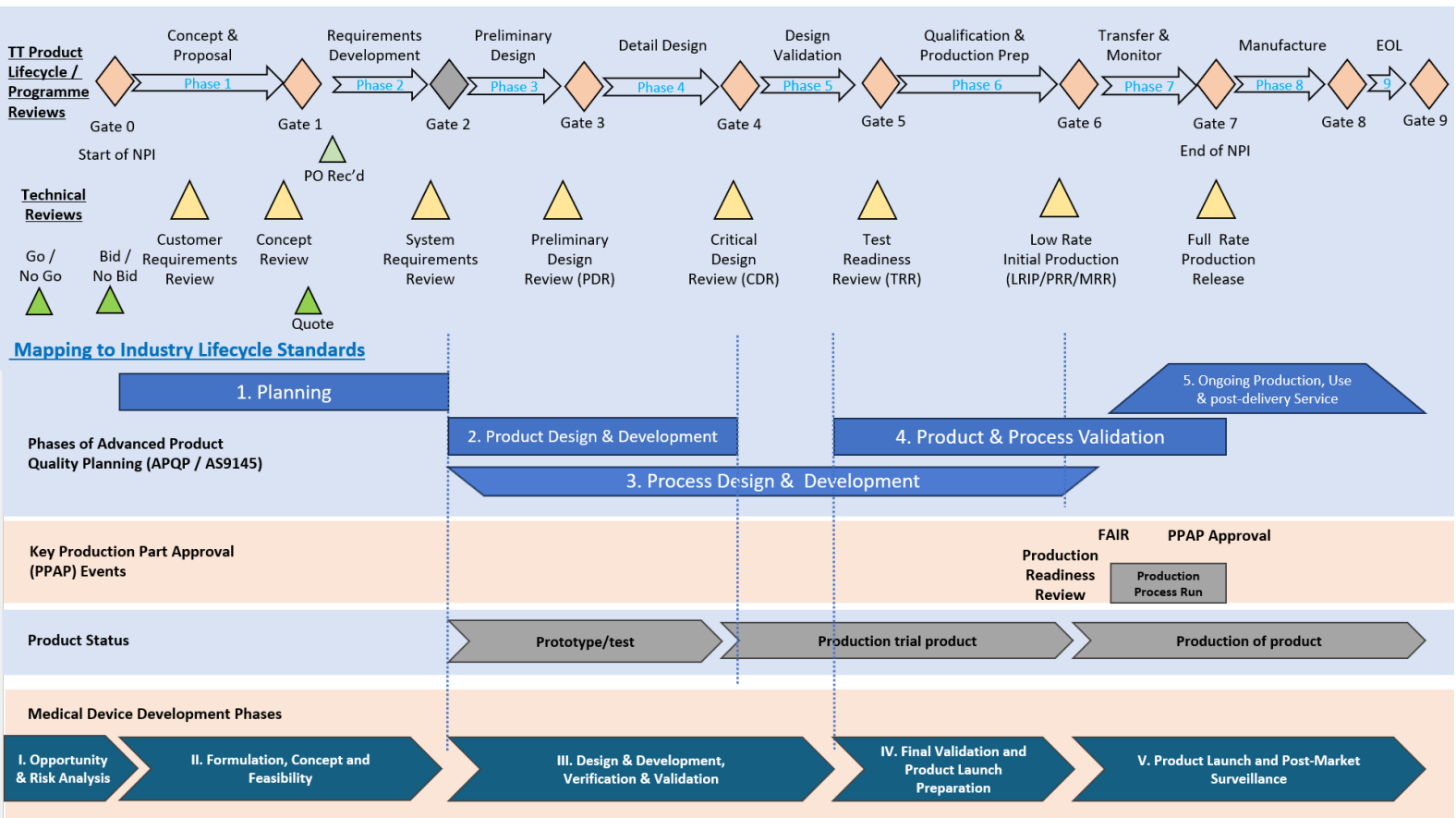


Figure 2.2.1 TT Product Lifecycle Alignment to Standards.

3 Project Types and Categories

Projects are classified in two ways, firstly (a) by the type, size, and complexity of the project, and secondly (b) by the revenue and business impact.

3.1 Project Types

The New Product Development (NPD) size classification relates to the templates that have been defined for a starting point for project scheduling. NPD and New Product Introduction (NPI) may be used inter-changeably to reflect terminology used at different locations.

3.1.1 NPI A (Large):

- Complex system or unit
- Typically, >4000 hours of engineering

3.1.2 NPI B (Medium):

- New component or less complex system / unit
- Typically, 500 - 4000 hours of engineering

3.1.3 NPI C (Small):

- Modification or variants of existing product.
- Typically, <500 hours of engineering

3.1.4 NPI D Minor Customisations (Momentum)

- Minor customization of an existing product to meet specific customer demand, or;
- Minor changes to an existing product to address material obsolescence, yield improvement or in-service issues, or;
- Small amount of engineering to support order of infrequently repeating product.
- Typically <40hrs but not expected to be >60hrs (otherwise NPI C or Continuous Improvement as appropriate).

3.1.5 Make-to-Print

Project where the customer owns the design, with minimal or no input to the design process from TT, other than advising on manufacturability. Synonym Build-to-Print may be used at some locations.

3.1.6 NTI: New Technology Introduction

Projects that develop a new product or manufacturing technology that is not necessarily unique to a single product development. These follow the industry standard maturity model "Technology Readiness Level". Many variations of this scale exist since it's invention by NASA, with definitions modified to align with the needs of specific sectors of industry. The TT Electronics definitions are based on a blend of these, tailored for the range of product types developed within our various businesses. For the purposes of our business and for the sake of not over-complicating, the Manufacturing Readiness Level is not considered separately, with the TT NTI process including considerations of manufacturing readiness. TRL, therefore can be considered to equal MRL i.e. TRL1 = MRL1.

TRL 1	Idea for a New Technology	Basic principles observed and reported – transition from scientific to applied research
TRL 2	Technology concept	Technology concept and/or application formulated – Applied research theory focused on specific applications
TRL 3	Technology proof of concept	Proof of critical characteristics through analytical and/or lab studies.
TRL 4	Technology basic validation in laboratory environment	Technology demonstrates basic functionality in a lab with associated performance predictions defined relative to the operating environment.
TRL 5	Technology basic validation in a relevant environment	Technology built into component or sub-system prototype with testing to demonstrate overall performance in critical areas.
TRL 6	Technology Demonstration in a representative environment	Technology constructed into a mature prototype with testing representative of the operational environment (pre-qual level testing).
TRL 7	Technology demonstration in an operational environment	Technology implemented in a product or process, demonstrated critical functionality in operational environment.
TRL 8	Technology qualified through test and demonstration	Product or process implementing technology has been qualified through test. Technology proven to work in its final form and under expected conditions.
TRL 9	Technology qualified through successful operations	Product or process implementing technology has been demonstrated to work through operational experience, with successful real-world performance.

Figure 3.1.1 TT Electronics Technology Readiness Level Scale

Internally executed NTI projects within TT will normally focus on TRL4-6, with TRL1-3 being the realm for university-based research and TRL 7-9 achieved through integration in a product or process that enters operation, therefore covered in NPI processes.

The workflow of an NTI project starts the same as a product development, with Gate 0 and Gate 1 having the same meaning, to generate a proposal whether internal or external. There is an additional review prior to Gate 1 to ascertain the starting level of TRL, normally expected to be TRL3. Once approved to continue past Gate 1, the following Gates review the maturity of the technology from TRL4 through to TRL6. This is illustrated in Figure 3.1.2.

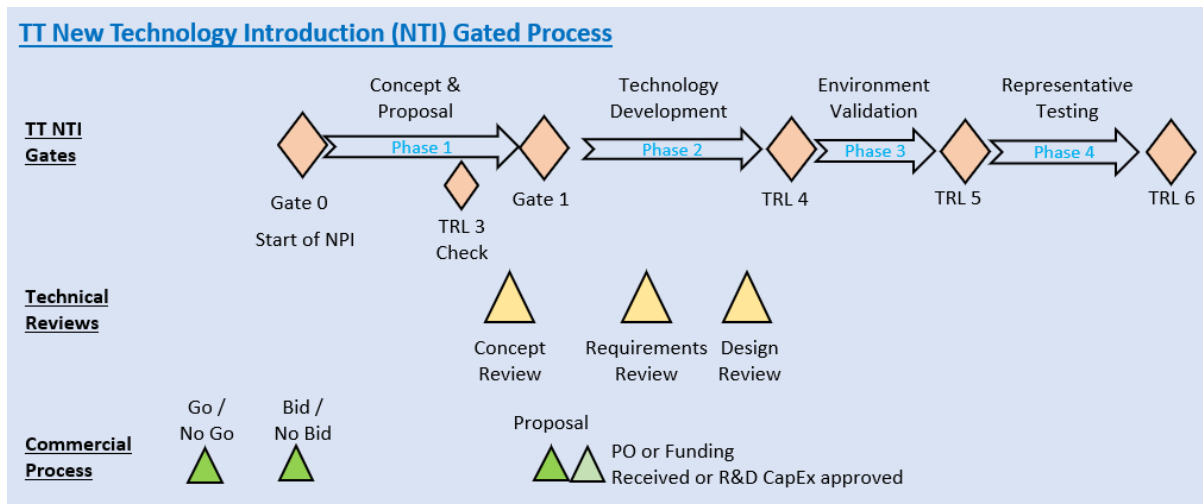


Figure 3.1.2 TT Electronics NTI Gated Process

Continuous Improvement Projects:

These may be controlled using similar mechanisms of Stages and Gates, but are not in scope of this procedure and are listed here only for completeness.

3.1.7 Continuous Improvement (CI) PDCA

PDCA = Plan, Do, Check, Act. This is normally appropriate for smaller, simpler improvement projects that are typically of short duration, or those that do not readily fit with being data driven. It may also be used for transfer projects (i.e. when product manufacture is moved from one location to another).

3.1.8 Continuous Improvement (CI) DMAIC

DMAIC – Define, Measure, Analyse, Improve, Control. Standard data driven structure used by six-sigma projects.

3.1.9 Continuous Improvement (CI) TT BE Lean

TT Electronics developed a form of a Lean Six-Sigma type project structure. This is not yet implemented in a standard template but may be used as a project structure for process improvement. In this case, select DMAIC type and map the Gates appropriately in the project plan.

3.2 Project Categorisation

The Project Category is defined by the revenue, TT investment and business impact / risk:

3.2.1 Cat 1

- Project linked with forecast total lifetime revenues of >£5m
- Any project requiring an internal (TT) investment (CapEx or OpEx) of >£100k
- All NPI A projects
- NTI project for strategically important new technology linked to significant revenues.

3.2.2 Cat 2

- Project linked with forecast total lifetime revenues of >£500k but less than £5m
- Any project requiring an internal (TT) investment (CapEx or OpEx) of >£50k
- All NPI B projects not qualifying in Cat 1
- Any NTI project not qualifying in Cat 1

3.2.3 Cat 3

Any project not fitting Cat 1 or Cat 2 definitions.

Following table summarises definitions of Category and mapping to which project Types are applicable;

	Cat 1	Cat 2	Cat 3
Total Revenues	>=£5m	>=500k & <£5m	<£500k
or TT Investment	>=£100k	>=£50k	<£50k
NPI A	All	No	No
NPI B	Yes	Yes	No
NPI C	Yes	Yes	Yes
Minor Custom	Yes	Yes	Yes
NTI	TT Strategic	All other	No
Make-to-Print	Yes	Yes	Yes
CI - PDCA	Not preferred	Not preferred	Yes
CI - DMAIC	Yes	Yes	Yes
CI – TT BE Lean	Yes	Yes	Yes

Figure 3.2 Category and Type

4 Gate Reviews

4.1 Gate Content and Structure

It is normally expected that the Gate Review will be a meeting with all the necessary stakeholders present. The review should consider the current status of the project, external factors impacting the business case, and ensure activities of the preceding Phase are complete.

It is expected that the review starts with a presentation of the background and status of the project. Best practice is that is done using a presentation that is retained as a record, including as a minimum:

- Description and illustrations to explain the product or technology being developed.
- Any changes of scope since project start.
- Financial summary of the project, including expected, actual and forecast costs and income (where relevant).
- Schedule, comparing baseline and actual/forecast.
- Any significant issues impacting the project.
- Update of key Risks and Opportunities.
- Business case update.

The outcome of a Gate Review can be:

- Pass – project can proceed to the next phase.
- Conditional Pass – project can proceed to next phase, but some actions must be addressed either before project proceeds or in parallel within a defined timeframe (which applies must be formally documented, as a list of the actions).
- Fail – project is either stopped or must complete actions and then re-hold the gate review. Which applies must be documented in the record of the review.

In cases of exceptional business need, a “Fail” Gate Review outcome can be over-ridden by the site lead / GM or relevant EVP to allow work to proceed in the next phase, provided this is formally documented and conditions of the project proceeding are captured.

The Gate exit criteria are listed in section 6.

Best practice:

- A defined checklist is used in each Gate Review to ensure all relevant topics are covered, actions captured, and outcome documented.
- Gate actions recorded in the project Action Register in PPM system (see section **Error! Reference source not found.**).
- Checklist stored in the PPM system with the Gate review record as auditable evidence. Gate review must be signed in PPM system as auditable record.

5 Project Portfolio Management

5.1 Tools & Systems

Every product and technology development project within the group is required to be managed within a structured system, maintaining auditable records enabling monthly reporting on the following as a minimum:

- An overview of the project, describing as a minimum what the project is about, why we are doing it, the scope, overall timeline, value (associated revenues), summary of costs and target customer(s) / markets.
- A scheduled task list / Gantt showing progress / % complete.
- Commentary on progress, including tasks complete undertaken in last period, deviations from the planned tasks / activities and changes in scope.
- Tasks planned for next period
- Costs incurred and forecast vs costs planned (both for labour and external costs)
- A register of key Risks, Opportunities & Issues
- Action register (also referred to as Rolling Action Item List - RAIL)
- Method of forecasting labour resource demand per project.

The current standard Project Portfolio Management (PPM) system used in TT Electronics is the Cora Systems PPM (Cora) and its use is recommended for all product and technology development projects in Europe and North America. Each site may determine a policy for how projects will be managed, however the minimum criteria for management and reporting above apply regardless of the tools and systems used.

Among the many benefits of using the provided standardised PPM system are:

- Provide a suite of tools to help project leaders deliver projects on time, quality and cost, through management of tasks, schedules, resources, risks, actions, and records / artifacts, within a fully auditable single environment.
- Have a “single source of the truth” for all data relating to the execution of projects. This is a critical enabler for increased use of GenAI business intelligence that will allow us to become more competitive.
- Understand the status of all projects across our portfolio, to make more informed priority decisions on where help is needed, and which projects to continue, accelerate, stop or hold.
- Improve understanding of the benefit of the investments being made in product development.
- Forecast the resources and skills needed to execute all projects across our portfolio, to support planning of recruitment and training.

Cora has been configured to help guide projects through the gated process defined in this procedure. Templates have been created that map directly to the standard project types, including a starting framework for the schedule of each type of project. By following the defined process steps, you are guided to create a project structure that will align with group policy and that will work with the tools and reports that are developed or will be developed.

This procedure is not intended to provide detail guidance of how to use the Cora system but provides a top-level framework to signpost you to what is expected and how to comply with group policy. The Cora system has a built-in help called Cora Assistant, accessible through the “?” in the top right of the Cora screen. Additional guides tailored for TT specific use are available in Cora through the top menu option Company->Documents, under the section Training Materials. The process flow defined in the procedure is also available in the Documents section of Cora. Additional training materials will be added over time, so you are encouraged to revisit this section if trying to do something you are not familiar with.

Every new product or technology project must be created in Cora. It is normally expected that one project is created for the entire lifecycle of a single product. For example, if responding to an RFQ or similar for a quote, create a project with the template for the whole project as if we are successful. You can then update the task list in Cora to help with costing of the project. When passing Gate 1 to submit the proposal, the project status feature (Project Properties menu) should be used to put the project on Hold to indicate that we are not actively working on it but are waiting on the customer PO. The project status should then be reset to In Progress if we get a PO or follow the Cancelled/Closed workflow under the Home menu where we do not win the work. This retains all auditable records of a product development in a single place and is much easier to maintain and trace.

5.2 Cora Functionality

Cora is a broad and highly configurable system, with a large array of functionality. The extent to which this functionality is used in an individual project is in part down to the project leader, however there are certain aspects which are mandatory. The following list of functionalities does not cover everything possible in Cora but some of the key features and fit with TT policy:

5.2.1 Mandatory (Europe and North America Regions)

- Creation of projects using the “ADD->Add Project” workflow (top left in main window). This provides a step-by-step wizard for requesting a new project to be created and should normally be initiated by the project leader. Sufficient detail must be included in the Executive Summary section of the form to allow someone not familiar with the project to understand the scope, including what product or technology the project will develop; who is the customer or what is the market; financial information such as cost of development; what resources are required (engineering hours, capital equipment, facilities, etc as relevant); overall schedule, including any critical deadlines.
- Moving projects through Gates and retaining record of approvals using the Phase Gates menu. Group policy is that the minimum approval in Cora for each Gate is the Project Manager and the lead Design Engineer. If only these approvals are being used then Project Manager is signing to affirm that they have received all necessary approvals for that Gate, which should be evidenced by attaching a record of the Gate Review (minutes, checklist or similar). Each site can optionally have a policy for including a maximum of three other approvers for each Gate, which should be documented in the site BMS / QMS, alongside this procedure.
- Gantt / Task List, including key milestones that must capture as a minimum the dates of every Gate Review (exception for Momentum type projects, where milestones can be reduced to the start and end of each minor product introduced). Each task in the Gantt must have a defined Phase that it relates to in the field of that name.

- Maintain Project status under Home->Properties menu
- Schedule->Utilities->Project Baseline. As a minimum, a Baseline must be created when the project is first loaded, when a proposal is submitted to a customer and again if a PO is received from a customer. In each case the Baseline must reflect the
- Complete and maintain mandatory fields (marked with red asterisk) under Home->Project Information Page.
- Promptly closing out a project using the Close/Complete Project workflow (Home menu) if project is either cancelled or completed.
- Ensure site is set to the lead site (and only the lead site) under Home->Properties menu, Organisations tab)
- Reporting – Health Check (update the project health to indicate status vs Time, Costs and Quality).

5.2.2 Recommended Functionality

- Gantt / Task List, using provided standard templates.
- Resourcing assigned by Skills. Every project should have the resource demand defined and maintained (under Planned Resources) against the project to reflect the current best view of forecast. At the time of a proposal or agreed KO of an internally funded development, this must reflect the resource (hours) included in the approved cost estimate.
- Resourcing assigned under Materials to include both material and subcontract spend. Every project must have bought out costs defined and maintained (under Planned Resources) to reflect the current best forecast. At the time of a proposal or agreed KO of an internally funded development, this must reflect all bought-out costs included in the approved cost estimate.
- Maintain an action list (RAIL) under Registers->Action Register
- Manage Risks and Issues under Register->Risks and Issues
- Defining deliverables on Milestones and creating Key Milestones in the Gantt/Task list so that the summary milestone Gantt works effectively.
- Set baselines (schedule->utilities) every month and at major milestone like significant technical reviews so you can track progression over time.
- Manage Decision through the Decision Register to provide an auditable trail.
- NB there is a known feature upcoming for a Change Register, which is strongly recommended to adopt once it is available.
- For larger teams, suggest selecting the options for sending email alerts to assigned resources (under Home->Properties menu)
- Add images to the Project Information Page to illustrate the project and progress (these will be picked up by the report below).
- Reporting – Snapshot Report using the “TT Project Review (system)” template. This provides a structure to standardise how we review projects and will be improved over time as functionality becomes available. To use it effectively you must modify the date range under the Milestones section, which requires that you have defined a Baseline, which should normally be the one associate to the schedule agreed for the currently agreed or contracted scope (often the baseline at KO).

- Reporting – Health Check : commentary + list of recent activities and for next period.
- Reporting – Resource Manager
- Portfolio Dashboards, from top menu, Dashboards (access dependant). Tools to explore multiple projects in a portfolio, aimed at management.

6 Gate Exit Criteria

The following tables present the exist criteria for each Gate in the development process. The following definitions apply:

- **Responsible:** Defines the function(s), person or people who normally perform the task in the preceding phase to provide the answer to the exit criteria question. Their agreement is needed to the answer given.
- **Accountable:** Function or individual who is ultimately answerable for the correct and thorough completion of the task that addresses the exit criteria. This person or function has the ultimate "ownership" of the task and ensures it gets done.

The last four columns indicate which criteria are either mandatory (M), optional (O) or informational (I) according to the complexity (Type) of the project (see [Project Types](#))

6.1 Gate 0 Exit

Responsible	Accountable	Question	A	B	C	D
Commercial	Commercial	Is opportunity entered in Salesforce including information on application, estimated value and timing?	M	M	M	O
Commercial	Commercial	Has a description of the opportunity been presented to the review with enough information for everyone to understand what it is?	M	M	M	M
Design Eng	Commercial	Do we have the technical capability to quote and execute the project? (if this is totally outside our capability then reject, otherwise discuss shortfall in GO and set actions to address if required)	M	M	M	M
Design Eng	Commercial	Do we have sufficient information (drawings, specs, etc) to be able to start to prepare a proposal or do we need more information from the customer?	M	M	M	M
Purchasing	Commercial	If applicable, do we believe it will be possible to get supplier quotes within the timeframe required for the bid/proposal? <i>Answer NA if project is not expected to include or require supplier quotes. As BOM is not defined at this stage, this is only an initial assessment to flag any potential issues</i>	O	O	O	O

All	Commercial	Does this fit within defined business strategy or is necessary due to current shorter-term commercial need?	M	M	M	M
All	Commercial	Do we have the capacity to respond to the quote in the timeframe required? (If not, add notes on how that could that be resolved and potential impact on other work)	M	M	O	I
All	Commercial	If the project is military, are security aspects understood and in place to allow us to work on the project?	M	M	M	M
All	Commercial	Do we understand any export controls that may we need to consider in preparing a quote? <i>Select NA if no specific export restrictions are expected to apply during quoting. Add comment in Notes if export control is potentially foreseen for resulting product.</i>	M	M	M	M

6.2 Gate 1 Exit

6.3 Gate 2 Exit

6.4 Gate 3 Exit

6.5 Gate 4 Exit

6.6 Gate 5 Exit

6.7 Gate 6 Exit

6.8 Gate 7 Exit

6.9 Gate 8 Exit

Gate 8 should be initiated following a decision to end-of-life a product (i.e. make it obsolete) and is intended as a kick-off of the end-of-life process as much as an exit from the prior phase (normal production).

NB Gate questions are not currently included in standard checklist; however, topics considered should include:

- Confirmation of the business case to make the product obsolete
- Planning for end-of-life, including customer communication as well as internal actions, assessment of time-line
- Planning for commercial aspect such as opportunities vs last time buys, potential for licencing to third parties or changing from make to buy approach for the product
- Capacity constraints vs end-of-life builds, etc. are understood
- Security and/or legal / contractual requirements for retained data, where relevant
- Plans for disposal of unused materials, including EHS and financial aspects

6.10 Gate 9 Exit

Gate 9 exit marks the End of Life of a product

NB Gate questions not currently defined in checklist; however, topics considered should include:

- Confirmation that all open end-of-life tasks / actions are closed
- Confirmation that all last-time-builds are complete, and all related contracts / POs closed
- Confirmation that ERP and other systems are appropriately updated to ensure we cannot take orders for the obsolete product
- Confirmation that any records and/or figs, fixtures are either appropriately destroyed or archived.
- Confirmation that any figs & fixtures are either appropriately destroyed or archived.
- Confirmation that remaining stocks of materials have been correctly accounted for or written off where the obsolete product is the only use for those materials